EX.NO: 10.

Date: 29/9/26 Internetworking with Routers in Cisco packet tracer simulator.

10)

a) Internetworking with router:

AIM:

To design and configure a simple internetwork using a router in cisco packet tracer and verify connectivity between two diffrent networks.

procedure:

1. Open Ciso Packet Tracer and create a new workspace.
2. place one Router and two pc's in the workplace.
3. connect PC0 to Router 1 FastEthernet 0/0 and PC1 to Router 1 FastEthert 0/1 using copper straight through cables.
4. open the router CLI and configure the two interface with Ip addrss:

   Fast Ethernet 0/0 :→ 192.168.10.1/24.

   Fast Ethert 0/1 :→ 192.168.20.1/24.

5. Enable Both interface using the no shutdown comand.
6. Configure PC0:

   Ip addrss: 192.168.10.2.

   Subnet mask: 255.255.255.0

   Default gway: 192.168.10.1

7. Configje PC1:

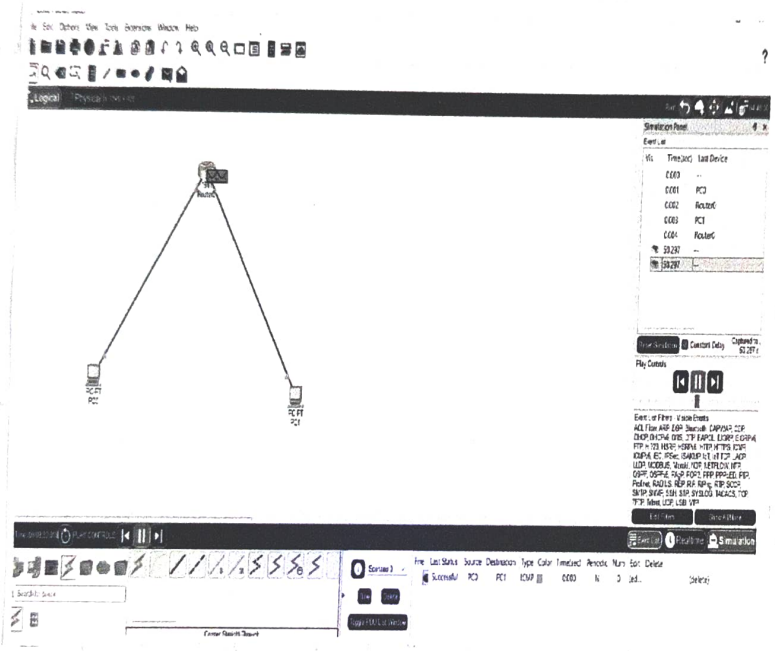
IP address : 192.168.20.2

Subnet mask : 255.255.255.0

Default gateway : 192.168.20.1

8. Switch to simulation mode and send a simple PDU from PC0 to PC1.

9. Observe the packet movement through the Router. and verify the acknowledgement from PC1



Result:

Thus, a simple internetwork using a router was successfully designed and configured in Cisco packet tracer, and connectivity between PC0 and PC1 was successfully verified using a simple PDU

10)

b) Internetwork using wireless Router, DHCP Server and Internet cloud.

AIM:

To design and configure an Internetwork using a wireless router, DHCP server and Internet cloud in cisco packet tracer.

procedure:

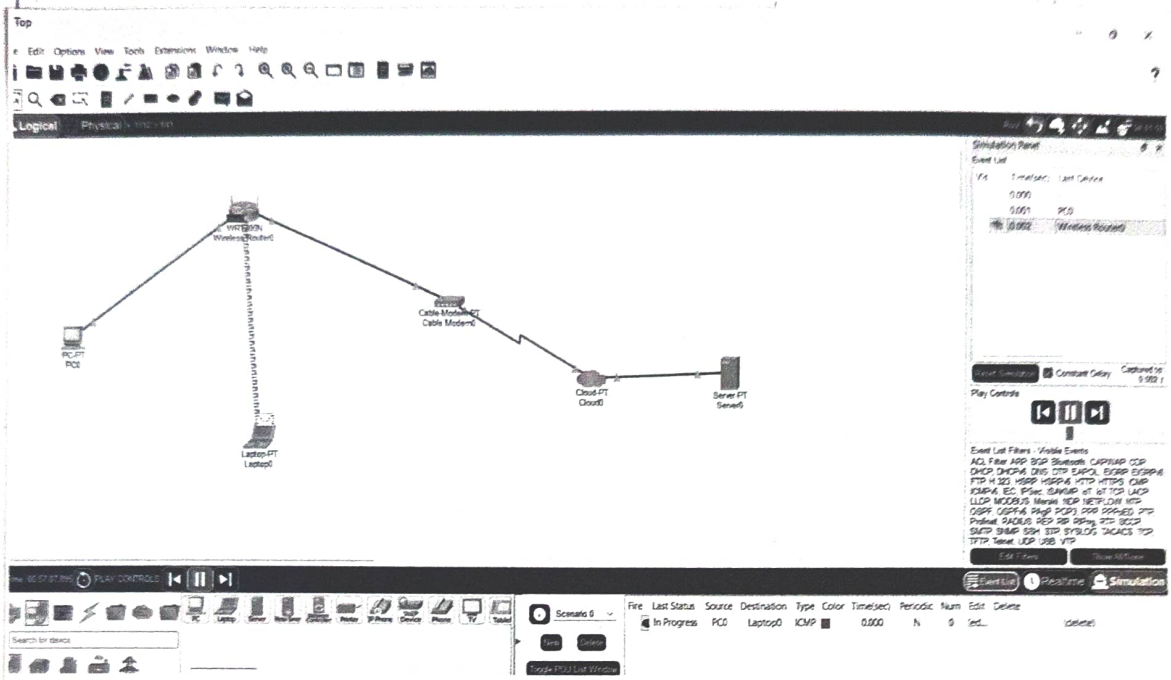
1. open cisco packet tracer and create a new work space.
2. place the following devices, wireless Router, PC, cable modem, Internet cloud, cisco.com server
3. connect.
   PC → wireless Router using copper straight through cable
   wireless Router → cable Modem using straight copper cable.
4. Internet cloud - cisco.com server using copper straight through cable.
5. open SSID as Homenetwork. and enable DHCP.
6. Configure the laptop with wpc 300N wireless module.
7. Configure the DNS as 208.67.220.220 and cisco.com.
8. on the PC, open:

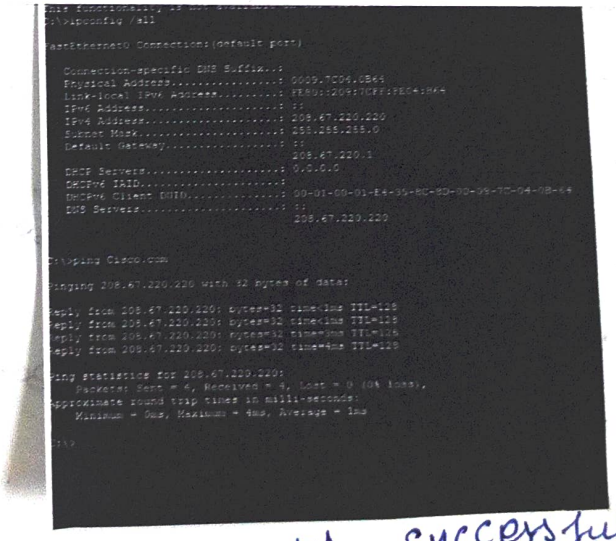
ipconfig /release

ipconfig /renew

9. Test connectivity using ping cisco.com.

10. Verify that replies are received successfully.





Result:

Thus, the wireless internetwork was successfully configured using a wireless router, DHCP server, Internet cloud, DNS server, and connectivity was verified using ping.